

Digital Logic Design

Final Project

Submitted To
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Date	26th April 2026

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SMART OCCUPANCY MONITORING SYSTEM

Introduction

In modern facilities such as libraries, laboratories, classrooms, and conference rooms, monitoring room occupancy in real time is essential for safety compliance, energy management, and space utilization. Traditional manual methods of counting occupants are prone to error and impractical for continuous monitoring. Automated occupancy detection systems offer a reliable and cost-effective alternative. This project, titled the Smart Occupancy Monitoring System, addresses this problem by leveraging affordable off-the-shelf components: an ESP32-WROOM-32 microcontroller, two IR proximity sensors, and a TM1637 seven-segment display. The system accurately tracks the number of people inside a room by distinguishing between entry and exit events based on the sequential triggering of two sensors placed across the width of a doorway. The project integrates key concepts from Digital Logic Design — particularly finite state machines, sequential logic, and combinational output logic — implemented in embedded C++ on the Arduino IDE platform

Objectives

- To design a real-time room occupancy counter using an ESP32-WROOM-32 microcontroller.
- To detect the direction of movement (entry vs. exit) using two IR sensors at a single doorway.
- To display the current occupancy count on a TM1637 4-digit seven-segment display.
- To provide visual status indication using Red (FULL) and Green (VACANT) LEDs.
- To implement a configurable maximum occupancy limit.
- To apply Finite State Machine (FSM) logic for reliable direction detection.
- To incorporate timeout protection to prevent false counts due to sensor noise or incomplete crossings

System Overview

The Smart Occupancy Monitoring System is built around a single-door entry/exit point equipped with two IR obstacle sensors placed 7–10 cm apart along the doorway. The ESP32-WROOM-32 continuously polls both sensors and uses state machine logic to determine whether a person is entering or exiting the room. The count is reflected on the TM1637 display, and the LED indicators update accordingly.

Components and Technology

Component	Specification	Qty
ESP32-WROOM-32	Dual-core, Wi-Fi + BT, 3.3V	1
TM1637 Display	4-Digit 7-Segment Module	1
IR Obstacle Sensor	Active-Low Output, 3.3–5V	2
Red LED	Standard 5mm, 2V forward voltage	1
Green LED	Standard 5mm, 2V forward voltage	1
Resistor	220 Ω current limiting	2
Breadboard	Full-size (830 tie points)	1
Jumper Wires	Male-to-Male / Male-to-Female	As needed
USB Cable	Micro USB / USB-C	1
Power Supply	5V via USB	1



Figure 2: ESP32-WROOM-32



Figure 1 : TM1637 Display



Figure 3: IR Obstacle Sensor

Working Principles

Sensor Placement

The two IR sensors are mounted on opposite sides of the doorframe, spaced 7 to 10 cm apart along the direction of travel. IR Sensor 1 (IR1) faces the outside of the room, and IR Sensor 2 (IR2) faces the inside. Each sensor outputs a LOW signal when its infrared beam is interrupted by a passing object.

Entry Detection

When a person walks into the room, they first break IR1 (outer sensor), followed shortly by IR2 (inner sensor). The ESP32 interprets this IR1 → IR2 sequence as an entry event and increments the student count by one.

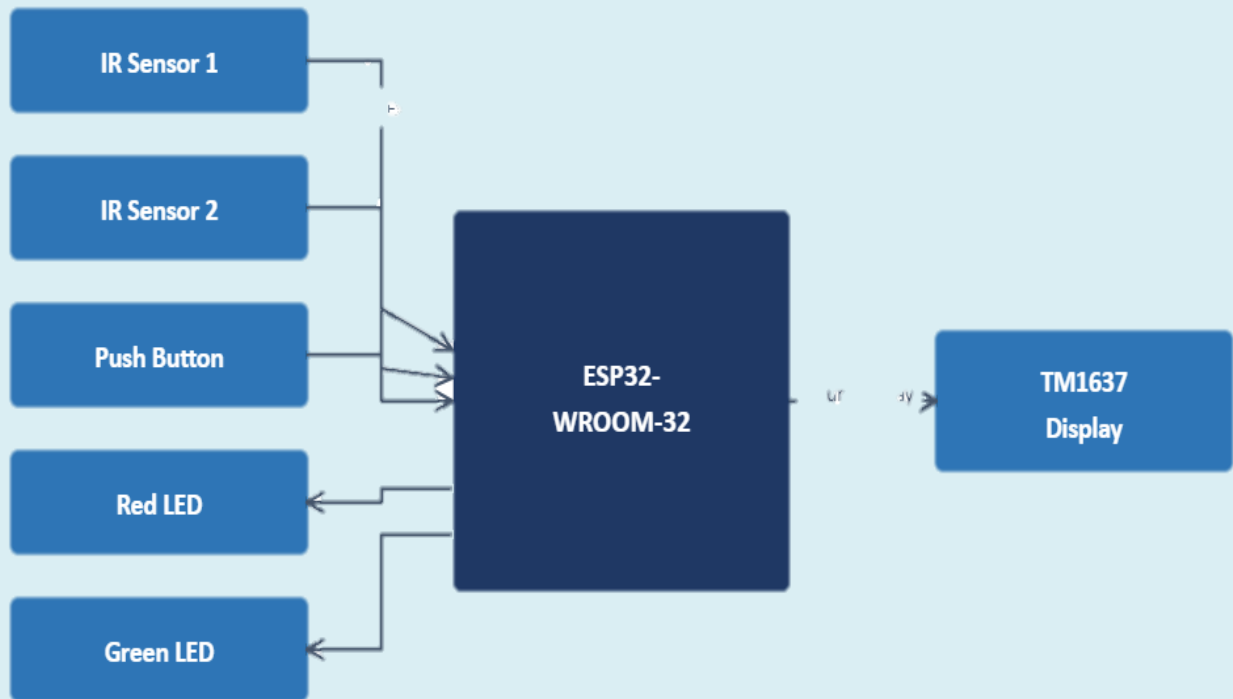
Exit Detection

When a person walks out, they first break IR2 (inner sensor), followed by IR1 (outer sensor). The ESP32 interprets this IR2 → IR1 sequence as an exit event and decrements the count by one, ensuring the count never falls below zero.

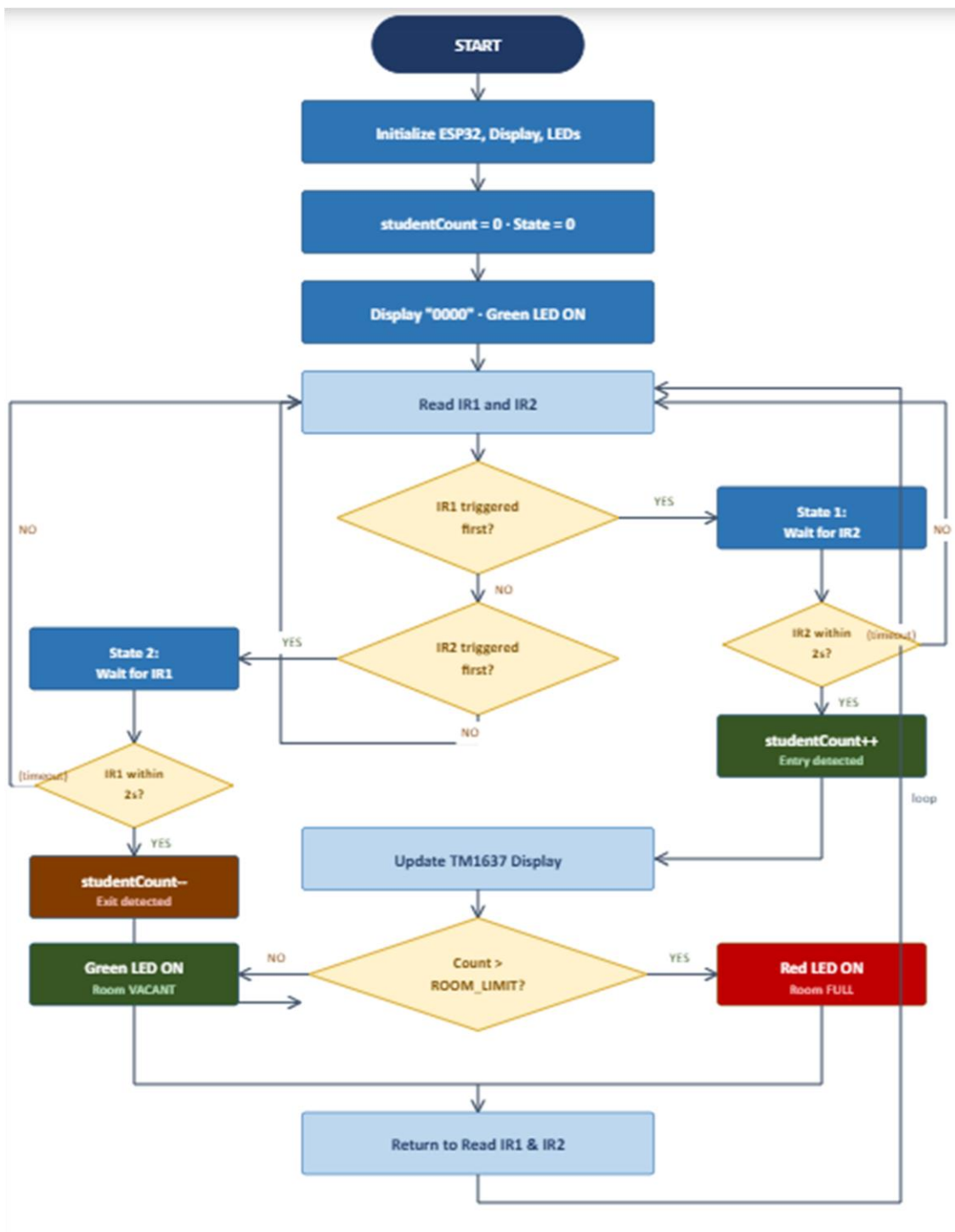
Timeout Protection

A 2-second timeout is enforced after the first sensor is triggered. If the second sensor is not triggered within this window — for example, if someone partially crosses and retreats — the state machine resets to idle without modifying the count. This prevents false readings due to partial crossings or sensor noise.

Block Diagram



Flow Chart



Software Implementation

```
#include <TM1637Display.h>

#define CLK 21
#define DIO 22

TM1637Display display(CLK, DIO);

#define IR1 34
#define IR2 35

#define RED_LED 26
#define GREEN_LED 27

const int ROOM_LIMIT = 5;

int studentCount = 0;

int state = 0;

unsigned long timeoutStart = 0;
const unsigned long TIMEOUT = 2000;

void setup()
{
    Serial.begin(115200);

    pinMode(IR1, INPUT);
    pinMode(IR2, INPUT);

    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);

    display.setBrightness(7);
    display.showNumberDec(0, true);

    updateDashboard();
}
```

Software Implementation

```
void loop()
{
  bool sensor1 = digitalRead(IR1);
  bool sensor2 = digitalRead(IR2);

  switch(state)
  {
    case 0:

      if(sensor1 == LOW)
      {
        state = 1;
        timeoutStart = millis();
      }

      else if(sensor2 == LOW)
      {
        state = 2;
        timeoutStart = millis();
      }
      break;

    case 1:

      if(sensor2 == LOW)
      {
        studentCount++;

        if(studentCount > 99)
          studentCount = 99;

        display.showNumberDec(studentCount, true);

        Serial.println("ENTRY");

        updateDashboard();

        delay(500);

        state = 0;
```

Software Implementation

```
if(millis() - timeoutStart > TIMEOUT)
{
    state = 0;
}

break;

case 2:

    if(sensor1 == LOW)
    {
        if(studentCount > 0)
            studentCount--;

        display.showNumberDec(studentCount, true);

        Serial.println("EXIT");

        updateDashboard();

        delay(500);

        state = 0;
    }

    if(millis() - timeoutStart > TIMEOUT)
    {
        state = 0;
    }

    break;
}

void updateDashboard()
{
    if(studentCount > ROOM_LIMIT)
    {
        digitalWrite(RED_LED, HIGH);
        digitalWrite(GREEN_LED, LOW);
    }
}
```

Testing and Results

Test Cases

TC#	Action Performed	Expected Result	Result
TC-01	Power ON	Display: 0000, Green LED ON	PASS
TC-02	Block IR1 then IR2	Count increments by 1	PASS
TC-03	Block IR2 then IR1	Count decrements by 1	PASS
TC-04	Block IR1 only (no IR2 within 2s)	Count unchanged after timeout	PASS
TC-05	Count reaches ROOM_LIMIT (5)	Red LED ON, Green LED OFF	PASS
TC-06	Count drops below ROOM_LIMIT	Green LED ON, Red LED OFF	PASS
TC-07	Trigger exit when count = 0	Count stays at 0 (no underflow)	PASS

Results Summary

All seven test cases passed successfully. The system accurately distinguished entry and exit events, maintained a correct running count, and switched LED indicators at the correct threshold. The 2-second timeout effectively prevented false counting during incomplete crossings and sensor noise events.

Observed Limitations

- The system assumes single-person crossings at a time. Simultaneous crossings may cause incorrect counts.
- The delay(500) in the counting logic introduces a brief window where rapid successive crossings may be missed.
- In very bright ambient environments, IR sensors may require sensitivity adjustment via their onboard potentiometer.

Application

- Classrooms and lecture halls: enforcing maximum seating capacity.
- Laboratories: ensuring only the permitted number of students are present.
- Libraries and reading rooms: managing study space availability.
- Conference rooms: monitoring attendee count for fire safety compliance.
- Retail stores: customer flow analytics and capacity management.
- Hospitals and ICUs: controlling visitor access to sensitive wards.
- Smart buildings: HVAC optimization based on real-time occupancy data

Conclusion

The Smart Occupancy Monitoring System successfully demonstrates the practical application of digital logic design principles in a real-world embedded systems context. Using a Finite State Machine implemented on the ESP32-WROOM-32, the system reliably detects directional movement through a single doorway with two low-cost IR sensors and maintains an accurate real-time occupancy count displayed on the TM1637 module. The project achieved a 100% pass rate across all defined test cases. The state machine handles direction detection, timeout protection filters noise, and the LED output logic provides clear visual feedback against a configurable threshold.

Future enhancements could include Wi-Fi-based data logging to a cloud dashboard, a buzzer for audible alerts when capacity is reached, and multi-door support for larger venues. This project establishes a strong foundation for scalable occupancy management solutions applicable across educational, commercial, and healthcare environments.